

Random Forest Model for Market Volatility Prediction

We built a machine learning model to classify market regimes into high volatility and low volatility based on technical indicators derived from SPY ETF historical data from the past 20 years. The overall process involved downloading historical SPY and VIX data using the `yfinance` package, then engineering a set of features including the rolling mean of returns, rolling standard deviation, rolling skewness and kurtosis, 20-day historical volatility (HV20), Average True Range (ATR), and the Relative Strength Index (RSI). To create the target variable, we labeled each day as high or low volatility depending on whether the day's absolute return exceeded the median of all historical absolute returns. Before training models, we applied feature scaling using `StandardScaler` to normalize the data.

We trained three different machine learning models: Random Forest, Logistic Regression, and K-Nearest Neighbors (KNN). Each model underwent hyperparameter optimization using `GridSearchCV` with 5-fold cross-validation to ensure fair and consistent evaluation. The Random Forest model, after optimization, achieved its best performance with a maximum depth of 5 and 100 estimators. On the test set, the Random Forest achieved an accuracy of approximately 66.04 percent, a precision of 66.25 percent, and a recall of 65.21 percent. Its AUC score was 0.731, indicating a reasonable ability to discriminate between high and low volatility periods.

When comparing models, Logistic Regression very slightly outperformed Random Forest in test accuracy, reaching around 67.7 percent, but Random Forest still demonstrated stronger AUC performance and better interpretability through feature importance analysis. K-Nearest Neighbors performed slightly worse than both Logistic Regression and Random Forest, with around 65.5 percent accuracy. Feature importance from the Random Forest model revealed that VIX closing values, annualized 20-day historical volatility (HV20), the RSI, and the rolling standard deviation were the most important predictors for identifying future volatility regimes. The strong influence of VIX and HV20 confirmed that external volatility measures and realized volatility remain critical signals for understanding regime shifts.

Overall, the Random Forest classifier proved effective for detecting volatility regimes, capturing both the complexity and the nonlinearity of the data. While Logistic Regression and Random Forest have almost the same performance, at this point the overall structure of the models matter more for us to make a preference. Future improvements could include adding macroeconomic indicators or using more advanced ensemble methods like XGBoost or LightGBM to push performance even further and developing trading strategies upon the model's predictions.